

Metamodel of Course Management (Textual Description)

A course management metamodel defines the structure and relationships between the core concepts required to represent and manage educational courses in an academic environment. It describes the elements that any course management system should support and how these elements interact.

1. Course

The Course is the central entity. It represents an academic subject offered by an institution. A course typically includes attributes such as course title, course code, description, duration, credit value, and learning objectives. A course is composed of multiple modules and is delivered by instructors to enrolled students.

2. Module

A Module represents a subdivision of a course that organizes content into manageable learning units. Each module may include topics, learning materials, assignments, and assessments. Modules help structure the learning path within a course.

3. Instructor

An Instructor is responsible for designing course content, delivering lectures, creating assignments, and evaluating student performance. An instructor can teach one or multiple courses, and a course may involve multiple instructors.

4. Student

A Student is a learner enrolled in one or more courses. Students interact with course materials, participate in activities, submit assignments, and complete assessments. Student performance is tracked throughout the course lifecycle.

5. Enrollment

Enrollment represents the relationship between students and courses. It records when a student joins a course and may include status information such as active, completed, or withdrawn.

6. Assignment

An Assignment is a task provided within a module or course that students must complete. Assignments may include submission deadlines, grading criteria, and feedback mechanisms.

7. Assessment

Assessment represents evaluation mechanisms such as quizzes, exams, or projects. Assessments measure student understanding and contribute to final grades.

8. Learning Material

Learning Materials include lecture notes, videos, presentations, documents, and other educational resources associated with modules or courses. These materials support student learning activities.

9. Grade

A Grade represents the evaluation result of assignments and assessments. Grades are associated with students and reflect their performance in a course.

10. Schedule

Schedule defines the timing of lectures, assignments, and assessments. It helps coordinate course delivery across instructors and students.

Relationships Summary

- A Course contains multiple Modules.
- A Course is taught by one or more Instructors.
- A Student enrolls in a Course through Enrollment.
- A Module includes Learning Materials and Assignments.
- Assignments and Assessments produce Grades.
- A Schedule organizes the timeline of course activities.

This metamodel provides a structured foundation for designing and implementing course management systems in educational platforms.

